Def: A ring R is called local if the set of nonunits of R forms an ideal of R.

Thm. TFAE for a ring R.
(1) R is local
(2) The set of nonunit of R coincides with $J(R)$, that is, $J(R)$ is the unique maximal left (or right) ideal of R.
(3) $R/J(R)$ is a division ring

Pf: Let M be the set of nonunits of R.
$(1)\Rightarrow(2)$ If M is an ideal of R, M contains all proper left ideals of R, then M is the maximal left ideal of R
$(2)\Rightarrow(3)$. If $a\in R\setminus J(R)$, then a is a unit.
So $\bar{a}\in R/J(R)$ is a unit.
$(3)\Rightarrow(1)$ If $a\notin J(R)$, then $\bar{a}\in R/J(R)$ is a unit.
So $\exists b\in R$, $c\in J(R)$ st $ba=1-c$ $\Rightarrow$ ba has a left inverse, so does a.
Similarly a has a right inverse $\Rightarrow$ a is a unit
So $M=J(R)$ is an ideal

Thm: If R is a local ring, then it has no idempotent other than the identity.

Pf: If R has a idempotent $e\neq 1$.
then both e and $1-e$ are nonunits since $e(1-e)=0$
So $e, 1-e\in J(R)$ $\Rightarrow$ $1=e+(1-e)\in J(R)$ contra

$v(ab)=v(a)+v(b)$
$v(a+b)=\min\{v(a),v(b)\}$
$R=\{a\in K\mid v(a)\geq 0\}$ is a subring of K $\rightsquigarrow$ called the valuation ring of v.
Let $R^{-1}=\{a^{-1}\mid a\in R, a\neq 0\}$ $\rightsquigarrow$ $K=R\cup R^{-1}$ $\rightsquigarrow$ So K is the fractional field of R.
(K,v) valuation field.
Let $P=\{a\in K\mid v(a)>0\}$ is an ideal of R called the valuation ideal of v.
The group of units of R: $R^{\times}=\{a\in R\mid v(a)=0\}$.

$\rightsquigarrow$ R is a local ring with $J(R)=P=\{\text{nonunits of } R\}$.
The field $\bar{R}=R/P$ $\rightsquigarrow$ called residue field of v.

Let $\lambda \in \mathbb{R}$. Define v' by $v'(a) = \lambda v(a)$, $\forall a \in K^\times$.

Then v' is also a valuation of K.

We say v is equivalent to v', write $v \sim v'$.

They give the same valuation ring, valuation ideal, group of units.

A valuation v of K is called discrete if its value group is isomophic to $\mathbb{Z}$.

If $v(K^\times) = \mathbb{Z}$, v is said to be a normalised discrete valuation.

Thm: (K, v) normalized discrete valuation.

R: valuton ring. $P = J(R)$.

(1) Let $\pi \in R$ with $v(\pi) = 1$. Then any $\alpha \in K$ is expressed as $\alpha = \pi^i u$. where $i = v(\alpha)$, $u \in R^\times$

(2) Any nonzero ideal of R is of the form $P^i = (\pi^i)$ for some $i \geq 0$

In particular R is a PID.

(3) P is integral closed [by (2)].

Pf: (1) If $v(\alpha) = i$, then $v(\pi^{-i}\alpha) = 0 \Rightarrow \pi^{-i}\alpha \in R^\times$.

(2) I be a nonzero ideal of R.

Let $i = \min\{v(a) \mid a \in I\}$.

Let $a \in I$, $a \neq 0$. $v(a) = j \geq i$

Then $a = \pi^j u = \pi^i (\pi^{j-i} u) \in \pi^i \Rightarrow I = (\pi^i)$

(for some $u \in R^\times$)

In particular, $P = (\pi)$ $I = P^i$

Cor. Two discrete valuations v and v' of K are equivalent iff they have the same valuation ideal.

$R = K \setminus P^{-1}$

$P^{-1} = \{a^{-1} \mid a \in P, a \neq 0\}$.

Thm discrete valuation ring = local PID.

Pf: only prove "$\Leftarrow$" Let R is a local PID. $J(R) = P = (\pi)$.

claim $\bigcap_{i=1}^{\infty} P^i = 0$

Let $0 \neq a \in \bigcap_{i=1}^{\infty} (\pi^i)$ then $a = \pi a_1 = \pi^2 a_2 = \cdots = \pi^i a_i$

So $\pi a_{i+1} = a_i \Rightarrow (a_1) \subsetneq (a_2) \subsetneq \cdots$ impossible as R is Noetherian

Then for any $a \in R \setminus \{0\}$, there exists an integer $i \geq 0$ s.t $a \in P^i$, $a \notin P^{i+1}$.

Then $a = \pi^i u$ with $u \in R^\times = R \setminus P$. Set $v(a) = i$

Let K be the fractional field of R. $v(b/a) = v(b) - v(a)$

v: normalized discrete valution.

(K, v) valuation field.

Fix $\gamma \in \mathbb{R}$ with $0 < \gamma < 1$.

Define $\varphi: K \to \mathbb{R}$ $\varphi(a) = \gamma^{v(a)}$

Then $\varphi(a) \geq 0$ $\varphi(a) = 0 \Leftrightarrow a = 0$.

$\varphi(ab) = \varphi(a)\varphi(b)$

$\varphi(a+b) \leq \max\{\varphi(a), \varphi(b)\}$

Exercise: $\varphi(a) = \varphi(-a)$. $\varphi(a^{-1}) = \varphi(a)^{-1}$

$\varphi(a) < \varphi(b)$. $\varphi(a+b) = \varphi(b)$

valuation ring $R = \{a \in K \mid \varphi(a) \leq 1\}$

ideal $P = \{a \in K \mid \varphi(a) < 1\}$.

conversely

If we have a function

$\varphi: K \to \mathbb{R}$.

satisfying (1), (2), (3)

Define $v(a) = \log_\gamma \varphi(a)$

$\Rightarrow v$ is a valuation of K

φ is called a multiplicative valuation of K.

$\varphi(K^\times)$ is a multiplicative subgroup of $\mathbb{R}^\times$ $\to$ value group of φ.

Two multi valuation φ_1, φ_2 are called equivalent if

$\exists\, \gamma \in \mathbb{R}, \gamma \geq 0$, s.t $\varphi_2(a) = \varphi_1(a)^\gamma$ $\forall a \in K$.

Define $d(a,b) = \varphi(a-b)$. $\to$ a metric on K.

K is a Hausdorff space.

A sequence $(a_n)=(a_1, a_2, \cdots a_n, \cdots)$ of elements of K is called a Cauchy sequence if $\forall \varepsilon>0, \exists N$ s.t $d(a_m, a_n)<\varepsilon$, $\forall m,n \geq N$

(a_n) is said to converge on an element $a \in K$ if $\lim\limits_{n\to\infty} d(a_n, a)=0$
we write $\lim\limits_{n\to\infty} a_n = a$. $\lim\limits_{n\to\infty} \varphi(a_n - a)$

If every cauchy sequence in K converges, K is called complete or a complete field, the associated valuation ring is said to be a complete valuation ring.

Given a valuation field (K, φ).
Let C be the set Cauchy sequence in K.
Define $(a_n)+(b_n)=(a_n+b_n)$ $(a_n)(b_n)=(a_nb_n)$

Then C is a commutative ring.
If I denotes the subset of C consisting of the sequene converging on 0.
I is a maximal ideal of C. (exicise)
So $\hat{K}=C/I$ is a field.

If (a_n) is a Cauchy sequence in K, then $|\varphi(a_m)-\varphi(a_n)| \leq \varphi(a_m - a_n)$
$\rightarrow (\varphi(a_n))$ is a Cauchy sequence in $\mathbb{R}$ $\rightarrow \lim\limits_{n\to\infty} \varphi(a_n)$ exists
If $(a_n)-(b_n) \in I$, then $\lim\limits_{n\to\infty} \varphi(a_n) = \lim\limits_{n\to\infty} \varphi(b_n)$
Define $\hat{\varphi}: \hat{K} \to \mathbb{R}$, $\hat{\varphi}(\widehat{(a_n)}) = \lim\limits_{n\to\infty} \varphi(a_n)$ $\widehat{(a_n)} = (a_n)+I$

Check $\hat{\varphi}$ is a multiplicative valuation of $\hat{K}$.
For $a \in K$, the sequence $(a)=(a, a, \ldots)$ is a Cauchy seq.
$\widehat{(a)} \neq \widehat{(b)}$ if $a \neq b$.

$K \subseteq \hat{K}$ $\hat{K}$ is an extension of K.
$a \mapsto \widehat{(a)}$
So $\hat{\varphi}$ is an extension of φ.
Lem $\hat{K}$ is complete (exercise)
So $\hat{K}$ is an extension of K s.t. $\hat{K}$ is complete and K is dense in $\hat{K}$
$\hat{K}$ is unique up to K-isomorphism of valuation fields
We call $\hat{K}$ the completion of K

Thm Let K be a valuation field with valuation ring R and valuation ideal P.
If $\hat{K}$ is the completion of K with valuation ring $\hat{R}$ and valuation ideal $\hat{P}$. Then $\hat{R}/\hat{P} \cong R/P$

Pf: clearly $R \subseteq \hat{R}$, $P = R \cap \hat{P}$. It suffices to show $R + \hat{P} = \hat{R}$. Let $0 \neq a \in \hat{R}$. Then $0 < \hat{\varphi}(a) \leq 1$
and $\exists a_n \in K$ s.t. $\lim_{n\to\infty} a_n = a$.
Hence $\lim_{n\to\infty} \hat{\varphi}(a_n - a) = 0$
So $\exists n$ st $\hat{\varphi}((a_n - a) + a) = \hat{\varphi}(a) \leq 1 \Rightarrow a_n \in R$.
So $a = a_n - (a_n - a) \in R + \hat{P} \Rightarrow \hat{R} = R + \hat{P}$.

normalized discrete valuation
valuation ring R $P = J(R)$.
Then $\{P, P^2, P^3, \cdots\}$ is a fundamental neighborhood system of 0

Therefore, (a_n) is a Cauchy sequence iff given a natural number l, there exists a natural number N st $a_m - a_n \in P^l$ whenever $m, n > N$

Also $\lim_{n\to\infty} a_n = a \Leftrightarrow$ given l, $\exists N$ s.t. $a_n - a \in P^l$ wherevers $n > N$.

Lem Let φ_1, φ_2 be discrete multiplicative valuations of K
TEAE (1) $\varphi_1 \sim \varphi_2$
(2) $\{a \in K \mid \varphi_1(a) < 1\} = \{a \in K \mid \varphi_2(a) < 1\}$
(3) The topologies on K induced by φ_1 and φ_2 are the same.

Pf: Let P_i be the valuation ideal of φ_i
(2) is $P_1 = P_2$. (1) $\Leftrightarrow$ (2)
(2) $\Rightarrow$ (3) $P_i^1, P_i^2, \cdots P_i^n \cdots$
(3) $\Rightarrow$ (2) $\varphi_1(a) < 1$. Then $\lim_{n\to\infty} \varphi_1(a^n) = \lim \varphi_1(a)^n = 0$.
$\lim_{n\to\infty} \varphi_2(a^n) = \lim_{n\to\infty} \varphi_2(a)^n = 0 \Rightarrow \varphi_2(a) < 1$

(K, φ) multiplicative valuation
V : a vector space over K.
A real-valued function $\|\cdot\|$ defined on V is said to be a norm on V if
(1) $\|v\| \geq 0$ and $\|v\| = 0$ iff $v = 0$
(2) $\|av\| = \varphi(a)\|v\|$ $\forall a \in K, v \in V$

(3) $\|u+v\| \le \max\{\|u\|, \|v\|\}$

If a norm $\|\cdot\|$ is given. $V \to$ metric space with $d(u,v) = \|u-v\|$.

$V \times V \to V$. $(u,v) \to u+v$
$K \times V \to V$. $(a, v) \to av$. } both continuous

Lemma: Let V be f.d. with a K-basis $u_1, \dots, u_n$.

For $v = a_1u_1 + \dots + a_nu_n \in V$, define $\|v\|_0$ by $\|v\|_0 = \max\{\varphi(a_i) \mid i = 1, \dots, n\}$.

This is a norm on V and the following hold.

(1) Let $v_r = a_{r1}u_1 + \dots + a_{rn}u_n \in V$. $r = 1, 2, \dots$

Then the sequence (v_r) is a Cauchy seq. iff the seq. $(a_{ri})_r$ in K is Cauchy seq. for all i

Also $\lim v_r = \sum_{i=1} a_i u_i \iff \lim_{r\to\infty} a_{ri} = a_i$ for all i

(2) K is complete

Thm Suppose K is complete. Then for any norm $\|\cdot\|$ on V there exists constants λ, μ such that $\|v\| \le \lambda \|v\|_0$ $\|v\|_0 \le \mu \|v\|$ for all $v \in V$.

Thus the two topologies induced by the two norms are the same.

Pf Let $\lambda = \max\{\|u_i\| \mid 1 \le i \le n\}$.

For $v = \sum_{i=1}^n a_i u_i$, $\|v\| \le \max\{\varphi(a_i)\|u_i\|\} \le \lambda \|v\|_0$.

We show the existence of μ by induction on n.

$n = 1$ ✓

$n > 1$ For $v = \sum_{i=1}^n a_i u_i$ set $\varphi_i(v) = \varphi(a_i)$

Claim: $\exists \mu_i$ s.t. $\varphi_i(v) \le \mu_i \|v\|$ for all $v \in V$.

(Then take $\mu = \max\{\mu_i\}$)

Let $V' = Ku_1 \oplus \dots \oplus Ku_{n-1}$

By induction $\Rightarrow$ $\|\ \|$ and $\|\ \|_0$ induce the same topo on V'.

V' is complete w.r.t $\|\cdot\|_0$ $\Rightarrow$ complete w.r.t $\|\cdot\|$ $\Rightarrow$ V' is closed in $V/\|\cdot\|$.

Suppose by the way of contradiction there is no μ_n with $\varphi_n(v) \le \mu_n \|v\|$ $\forall v \in V$.

For $j \in \mathbb{Z}, j > 0$ there is $v_j \in V \setminus V'$ st $\varphi_n(v_j) > j\|v_j\|$

This holds if v_j is replaced with non zero scalar multiple of it.

Hence we assume $v_j = a_{j1}u_1 + \dots + a_{jn-1}u_{n-1} + u_n$

$\Rightarrow \|v_j\| < \frac{1}{j}\varphi_n(v_j) = \frac{1}{j}$

$\lim_{n\to\infty} v_j = 0$ w.r.t. $\|\cdot\|$.

So $\lim_{j\to\infty}(v_j - u_n) = -u_n$

But $v_j - u_n \in V'$ for all j V' is closed in V.

So $u_n \in V'$ contradiction.

Let K. L be discrete valuation field with multiplicative valuations φ, Φ resp.
If $K \subseteq L$ and Φ is an extension of φ, we say (L, Φ) is an extension of (K, φ).

Thm, Let φ is complete discrete multiplication valuation of K and L is a finite extension of K.
If there exists an extension (L, Φ) of (K, φ) then Φ is unique up to equivalence.
Futhermore, (L, Φ) is complete.

R commutative ring
an R-algebra A over R.
$R \to Z(A)$